Rofhiwa Dagada exercise 4

(1) SELECT
e. EmployeeID,
e. first_name,
e. Last_name,
e. department,
e. Salary,
p. projectID,
p. projectName,
p. Status

from Employee. e
INNER JOIN Project. p
ON e. EmployeeID = p. EmployeeID;

EmployeeID

ID	first_name	Last name	Department	Salary	projectID	Project_name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	103	Cybersecurity Audit	75000	pending
4	David	Brown	IT	72000	104	Financial Analysis	90000	ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Michael	Clark	Finance	80000	106	Risk management	80000	pending

② SELECT

e.EmployeeID,
e.First_name,
e.Last_name,
e.Department,
e.Salary,
p.projectID,
p.projectName,
p.Status

from Employees.e
LEFT Join project.p
ON e.EmployeeID = p.EmployeeID

EmployeeID	first_name	Last_name	Department	Salary	projectID	projectname	Budger	Status
1	John	Doe	IT	70000	101	AI Developm	100000	completed
1	John	Doe	IT	70000	103	Cybersecurity	75000	pending
2	Alice	Smith	HR	60000	102	Employee Tra	50000	ongoing
3	Bob	Johnson	Finance	75000	104	financial Ana	90000	ongoing
4	David	Brown	IT	72000	null	null	null	null
5	Emma	wilson	Sales	65000	105	Market exp	65000	completed
6	Michael	Clark	finance	80000	106	Risk management	80000	pending

③ SELECT

p. ProjectID
p. ProjectName
p. Budget
p. Status
e. EmployeeID
e. firstname
e. Last name
e. Department
e. Salary

From Employees.e
Right Join projects.p
ON e. EmployeeID = p. EmployeeID

ProjectID	ProjectName	Budget	Status	EmployeeID	firstname	Lastname	Department	Salary
101	Ai Develop	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity	75000	pending	1	John	Doe	IT	70000
104	Financial	90000	ongoing	3	Bob	Johnson	finance	75000
105	Market	65000	Completed	5	Emma	Wilson	Sales	65000
106	Rust mar	80000	pending	6	Michael	Clark	Finance	80000

(4) SELECT

e. EmployeeID,
e. FirstName,
e. LastName,
e. Department,
e. Salary,
p. projectID,
p. projectName,
p. Budget,
p. Status

From Employee.e
Full OUTER JOIN projects.p
ON e.EmployeeID = p.EmployeeID

EmployeeID	FirstName	LastName	Department	Salary	PROJECTID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Developm	100000	Completed
1	John	Doe	IT	70000	103	cybersecurity	75000	pending
2	Alice	Smith	HR	60000	102	Employee trai	50000	ongoing
3	Bob	Johnson	Finance	75000	104	financial anal	90000	ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	market expan	65000	completed
6	Michael	Clark	finance	80000	106	Risk Management	80000	pending

(5) SELECT
City AS Location
from Employee
UNION
SELECT
Status AS Location
from projects

Location
New york
Los Angeles
Toronto
London
Sydney
Completed
ongoing
pending

(6) SELECT City AS Location
From Employees
UNION ALL
SELECT Status A Location
from project:

Location
New york
Los Angeles
Toronto
London
Sydney
New yort
completed
ongoing
pending
Ongoing
Completed
pending

(7) SELECT
EmployeeID,
first_name,
Last_name
Department,
Salary
from Employees
WHERE Salary > 70000;

EmployeeID	first_name	Lastname	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

(8) SELECT
EmployeeID,
first_name
Last_name
department
Salary
from Employees
WHERE Department IN ('IT', 'finance')

EmployeeID	first_name	Last_name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

(9) SELECT
ProjectID,
ProjectName,
Budget,
Status,
from projects
WHERE Status <> 'completed';

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	on going
103	Cyber Security Audit	75000	pending
104	financial Analysis	90000	ongoing
106	Risk management	80000	pending

(10) SELECT
ProjectID,
ProjectName,
Budget,
Status
From projects
WHERE Budget > 70000 AND Status <> 'Completed';

ProjectID	Projectname	Budget	Status
103	Cyber Security Audit	75000	pending
104	financial Analysis	90000	ongoing
106	Risk management	80000	pending

(11) SELECT
EmployeeID,
first name,
Last_name,
department,
Salary,
City
from Employees
WHERE City IN ('New york', 'Toronto')
ORDER By Salary DESC;

EmployeeID	first_name	Last_name	Department	Salary	City
6	Michael	Clark	finance	80000	New york
3	Bob	Johnson	finance	75000	Toronto
1	John	Doe	IT	70000	New york

(12) SELECT
EmployeeID,
first name,
Last name,
Department,
salary
from Employees
ORDER By Salary DESC
LIMIT 3;

EmployeeID	first_name	Last_name	Department	Salary
6	Michael	Clark	finance	80000
3	Bob	Johnson	finance	75000
4	David	Brown	IT	72000

(13) SELECT
Department,
Sum(Salary) As total_Salary
From Employees
Group By Department
ORDER By total_salary DESC;

Department	total_Salary
Finance	155000
IT	142000
Sales	65000
HR	60000

(14) SELECT City,
AVG(Salary) AS Average_salary
from Employees
group By City
Having Average_Salary > 65000;

City	Average_Salary
New York	75000
Toronto	75000
London	72000

(15) SELECT
department
COUNT(DISTINCT EmployeeID) AS number_of_Employees
FROM Employees
group By department
Having number_of_Employees > 1

department	number_of_Employees
IT	2
finance	2

(16) SELECT
status,
Count (DISTINCT projectID) AS number_of_projects
FROM ~~Employees~~ projects
group By Status
Having number_of_projects >= 2;

Status	number_of_projects
completed	2
ongoing	2
pending	2

(17) SELECT
e.EmployeeID,
e.first_name,
e.Last_name,
Sum(P.Budget) AS TotalProjectBudget
From Employees.e
INNER Join projects.P
ON e.EmployeeID = P.EmployeeID
Group By e.EmployeeID
HaVING Sum(P.Budget) > 150000;

output

EmployeeID	first name	Last_name	TotalProjectBudget
1	John	Doe	175000